

A Comprehensive L^AT_EX Document Example

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1 Introduction

This is an introduction to L^AT_EX features. L^AT_EX is great for writing technical and scientific documents.

2 Mathematical Equations

Here is an inline equation: $E = mc^2$.

A displayed equation:

$$\int_a^b f(x) \, dx = F(b) - F(a) \quad (1)$$

A matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad (2)$$

3 Tables and Figures

A simple table:

A	B	C
1	2	3
4	5	6

Table 1: A sample table

An example figure:

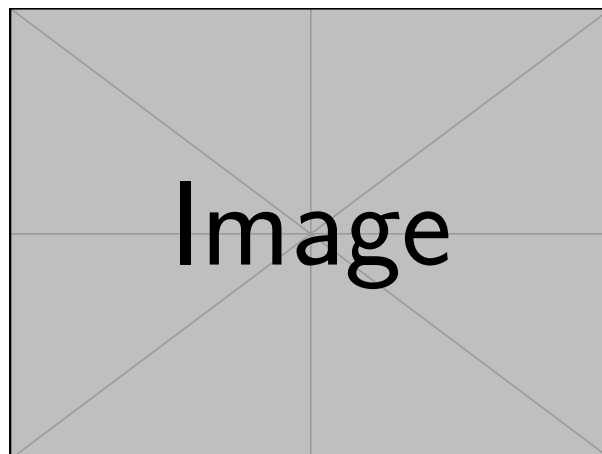


Figure 1: An example image

4 Code Listings

Here is some example code:

Listing 1: Example Python Code

```
def hello():  
    print("Hello , -World!")
```

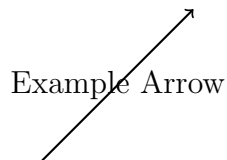
5 Algorithms

An example algorithm:

Algorithm 1 Example Algorithm

```
Initialize  $x = 0$   
for  $i = 1$  to  $n$  do  
     $x = x + i$   
end for  
Return  $x$ 
```

6 Diagrams with TikZ



7 Notations

Notations: Some common mathematical notations include:

- **Sets:** $\mathbb{N}$ (natural numbers), $\mathbb{Z}$ (integers), $\mathbb{R}$ (real numbers)
- **Operators:** $\sum_{i=1}^n i$, $\prod_{i=1}^n i$, $\lim_{x \rightarrow \infty} f(x)$
- **Logic:** $\forall x \in \mathbb{R}$, $\exists y \in \mathbb{R}$

8 Citations and References

We cite an example source [1].

References

- [1] John Doe and Jane Smith. “An Example Research Paper”. In: *Journal of Examples* 10.2 (2020), pp. 100–110. DOI: 10.1234/example.